IdiomX

A Large-Scale Bilingual Dataset for Idiomatic Expression Understanding

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1. Abstract

Idiomatic expressions pose a significant challenge in natural language processing due to their non-literal and context-dependent meanings. In this paper, we introduce IdiomX, a large-scale, high-quality bilingual dataset designed for idiom detection, interpretation, and cross-lingual understanding. The dataset contains over 123,000 contextualized examples derived from approximately 15,000 English idioms, each annotated with idiomatic and literal usage, semantic features, and English–Arabic translations.

To ensure quality and scalability, we employ a hybrid pipeline combining curated lexical resources with large language model (LLM)-based enrichment. The dataset undergoes rigorous validation, achieving over 99.98% label consistency and near-complete bilingual coverage. Additionally, semantic alignment metrics are introduced to quantify the consistency between idioms and their contextual usage.

IdiomX provides a comprehensive benchmark for idiomatic language understanding and enables research in multilingual and cross-lingual NLP. The dataset will be publicly released to support further advancements in idiom processing and semantic modeling.

To the best of our knowledge, IdiomX is the largest publicly available bilingual idiom dataset with contextualized examples and semantic annotations.

Keywords: idioms, NLP, bilingual dataset, semantic understanding, LLM, Arabic NLP

2. Introduction

Idiomatic expressions are a fundamental component of natural language, often conveying meanings that cannot be inferred directly from the individual words. This non-compositional nature makes idioms particularly challenging for natural language processing (NLP) systems, especially in tasks such as machine translation, semantic understanding, and information retrieval.

Despite their importance, existing idiom datasets are typically limited in size, lack contextual diversity, or are restricted to a single language. Furthermore, most available resources do not provide rich annotations such as semantic consistency, compositionality, or multilingual mappings.

To address these limitations, we present IdiomX, a large-scale bilingual dataset for idiomatic expression understanding. The dataset combines curated lexical sources with automated large language model (LLM) enrichment to generate high-quality contextual examples and annotations. In addition to English idioms, IdiomX provides aligned Arabic translations, enabling cross-lingual research.

Our contributions are as follows:

1. A large-scale dataset of idiomatic expressions with contextualized examples.
2. Balanced idiomatic and literal usage for robust modeling.
3. Near-complete English–Arabic bilingual coverage.
4. Semantic validation metrics to ensure dataset quality.
5. Rich linguistic annotations supporting advanced NLP tasks.

This work aims to provide a strong foundation for future research in idiomatic language understanding and multilingual NLP.

3. Related Work

Several datasets have been proposed for idiom detection and interpretation, including resources derived from lexical databases such as WordNet and Wiktionary. While these datasets provide useful foundations, they often lack contextual diversity and large-scale annotations.

Recent work has explored the use of neural models for idiom understanding, leveraging transformer-based architectures such as BERT and T5. However, the effectiveness of these models is often constrained by the availability of high-quality training data.

In multilingual settings, idiom datasets remain particularly scarce, with limited resources supporting cross-lingual analysis. This gap highlights the need for comprehensive datasets that combine scale, contextual richness, and multilingual support.

IdiomX addresses these challenges by providing a large-scale, bilingual dataset enriched with semantic annotations and validated using automated and rule-based techniques.

4. Dataset Construction

The IdiomX dataset is constructed through a multi-stage pipeline combining data collection, preprocessing, enrichment, and validation.

First, idiomatic expressions are collected from multiple lexical sources, including Wiktionary and WordNet. Duplicate entries and low-quality records are removed through normalization and filtering.

Next, contextual examples are generated and enriched using a large language model (LLM). For each idiom, multiple example sentences are produced, covering both idiomatic and literal usage. The enrichment process also generates English and Arabic meanings, as well as aligned translations of the example sentences.

To ensure consistency, additional linguistic features are computed, including compositionality level, learner difficulty, and semantic similarity scores. These features provide deeper insights into idiomatic behavior and support advanced modeling tasks.

The final dataset consists of over 123,000 examples with rich annotations and bilingual coverage.

The overall dataset construction pipeline is illustrated in Figure 1.

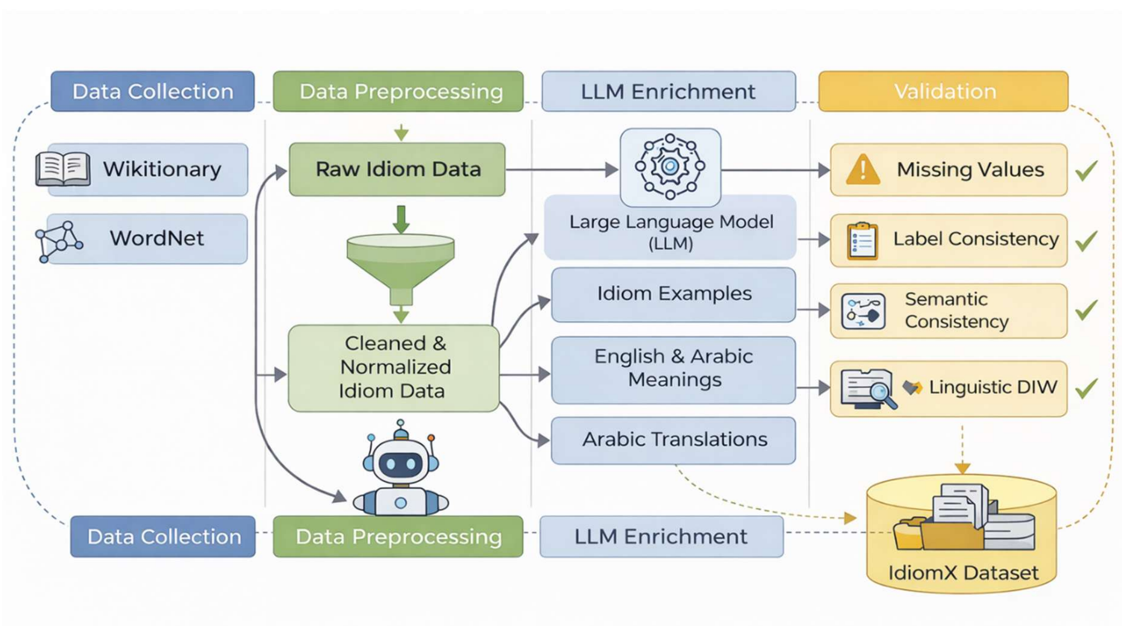


Figure 1: Overview of the IdiomX dataset construction pipeline

5. Data Validation & Quality

To ensure the reliability of the dataset, multiple validation steps are applied.

Missing values are analyzed across all fields, with critical attributes such as contextual examples and meanings exhibiting negligible missing rates (<0.01%). Label consistency between categorical annotations and boolean indicators exceeds 99.98%, indicating highly reliable labeling.

Surface-form alignment between idioms and their occurrences in context is evaluated using both strict and relaxed matching strategies, achieving high alignment rates.

Additionally, semantic consistency between idioms and their contextual usage is measured using embedding-based similarity metrics. The dataset achieves a mean semantic consistency score of 0.59, reflecting strong alignment while preserving natural variation.

These validation steps ensure that IdiomX is both high-quality and suitable for downstream NLP tasks.

6. Dataset Analysis

The IdiomX dataset contains approximately 14,986 unique idioms, with an average of 8.2 contextual examples per idiom. This provides both diversity and depth for modeling.

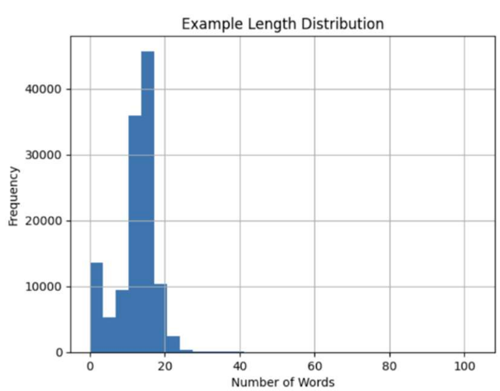
Example sentence lengths are well-balanced, with an average of approximately 12.5 words per example, ensuring sufficient contextual information without excessive noise.

The dataset is evenly balanced between idiomatic and literal usage, with each category representing approximately 50% of the data. This balance is critical for training robust classification models.

Furthermore, IdiomX provides near-complete bilingual coverage, with over 99.99% of examples and meanings available in both English and Arabic. This enables cross-lingual research and applications in multilingual NLP.

Overall, the dataset demonstrates strong statistical properties, linguistic diversity, and high-quality annotations.

Metric	Value
Total examples	123,336
Unique idioms	14,986
Avg examples/idiom	8.2
Arabic coverage	99.99%
Label balance	50/50



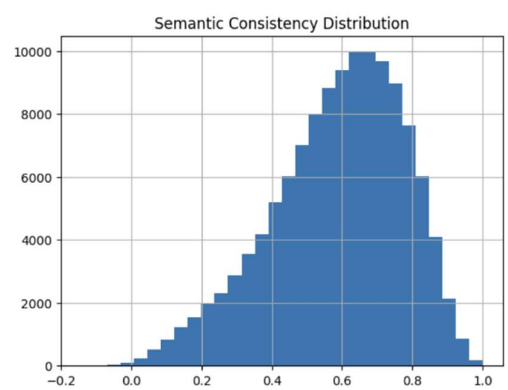


Figure 2: Distribution of example sentence lengths and semantic consistency.

7. Applications

The IdiomX dataset supports a wide range of natural language processing tasks, including:

- Idiom detection (idiomatic vs literal classification)
- Idiom interpretation and meaning retrieval
- Context-to-idiom generation
- Cross-lingual idiom translation
- Multilingual semantic understanding

The dataset is particularly valuable for evaluating transformer-based models and large language models in handling non-compositional language.

8. Dataset Availability

The IdiomX dataset will be publicly released via GitHub and Hugging Face to support reproducibility and further research. The repository includes the dataset, preprocessing scripts, and documentation for easy integration.

9. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding. By combining curated sources with LLM-based enrichment and rigorous validation, the dataset achieves high quality, strong semantic alignment, and near-complete bilingual coverage.

IdiomX addresses key limitations in existing resources and provides a comprehensive benchmark for idiom-related tasks. Future work includes extending the dataset to additional languages and exploring advanced modelling approaches using the dataset.

We release IdiomX to support further research in idiomatic language processing and multilingual NLP.

IdiomX establishes a new benchmark for bilingual idiomatic understanding and provides a foundation for future large-scale semantic modelling.